



GROUP PROJECT

College of Computing and Software Engineering

CSE 3153 – Database Systems

Project Part 1:

Part 1: Please read the business statement below and draw ER, NER diagrams. Submit two separate files of ER and NER diagrams. (100 points)

It is necessary for a technology company to have a database that can track their products and services. From the requirements for this database, the following information is collected:

1. A technology company offers its customers products and services.
2. Attributes of offering are offer ID and description.
3. Attributes of products are product name, standard price, and first release date.
4. Attributes of services are the name of the company's unit responsible for the service and service conditions.
5. There are also two different types of services: repair and maintenance.
 - 5.1.1: A repair service has a cost attribute and is the repair of some product.
 - 5.1.2: A maintenance service has an hourly rate attribute.
6. When a customer purchases an offering, the company will need to keep track of the date that the offering was purchased and the contact person for that offering.
7. Attributes of customers are customer ID, name, DOB, age, address, phone numbers, and email.
8. When a service is performed, that service is billed to some customer. Because some customers purchase offerings for their clients, a customer may be billed for services he or she did not purchase, as well as for ones that were purchased.
9. When a customer receives a bill for a service, the company must keep track of the date the service was performed, the bill due date, and the amount due.

Please note: any hand-drawn assignments of part 1 will not be accepted.

Project Part 2:

Part 2: Based on your NER diagram, create SQL tables and insert at least 3 records for each table. Write the SQL statements to answer the all questions below. Submit the SQL file and screenshots of running each query (one file including all the screenshots).

100 points (Create SQL tables & insert data: 30 points, Screenshots file: 10 points, SQL statements for the set of statistics: 60 points)

BA would like to have a set of statistics about the system as follows:

1. List of customer information and their purchases.
2. List of products under \$500.
3. List of the offerings and their products.
4. List of customers and the number of services they received.
5. What is the most expensive service and the name of the customer who received that service?
6. What is the oldest service and service information?